

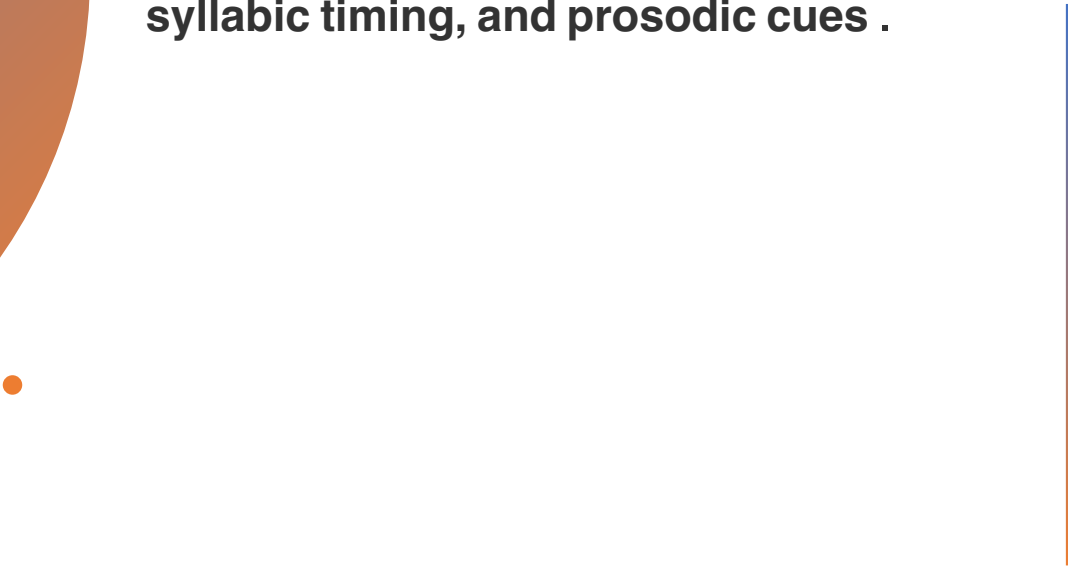
Other methods are
more prone to error and
arbitrariness.

Envelope Modulation Spectrum (EMS) as a measure of rhythm

more of an
automatic system.

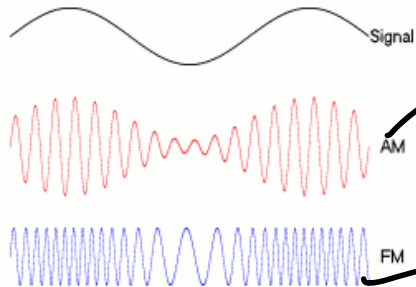


What is Amplitude Envelope?

- 
- The **amplitude envelope** represents how the "loudness" of a signal changes over time.
 - It captures intensity fluctuations, crucial for speech rhythm perception.
 - The envelope helps identify stress patterns, syllabic timing, and prosodic cues .
- 

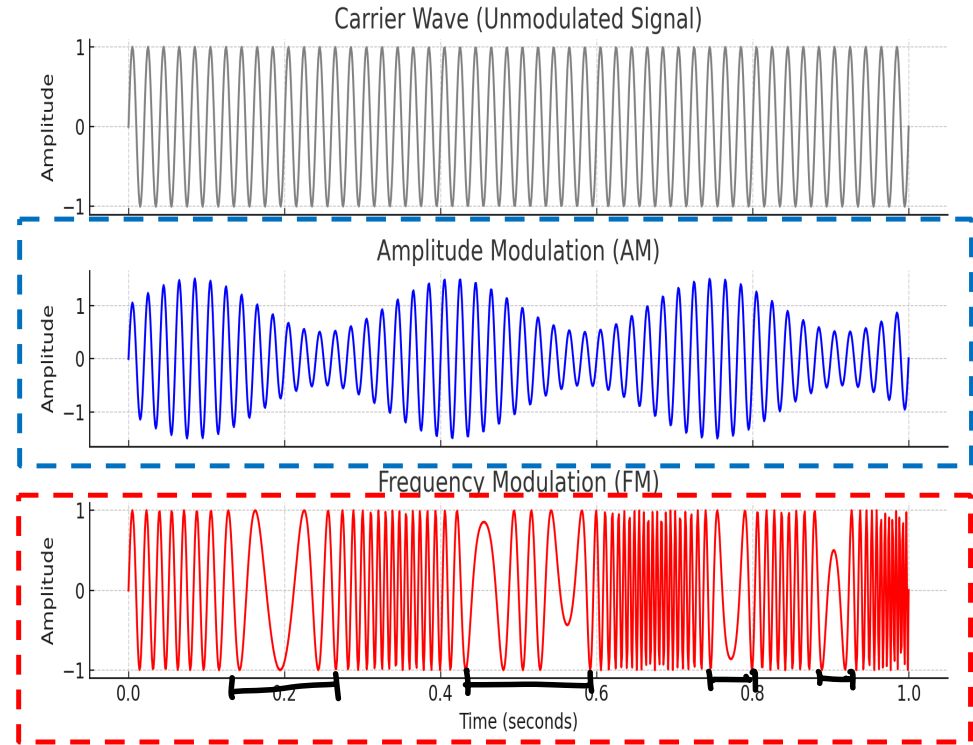
Amplitude Modulation vs. Frequency Modulation (FM)

- **AM (Amplitude Modulation):**
Changes in the amplitude envelope over time, influencing intensity fluctuations that contribute to rhythm and stress perception.
- **FM (Frequency Modulation):**
Changes in **pitch** over time (e.g., intonation contours).

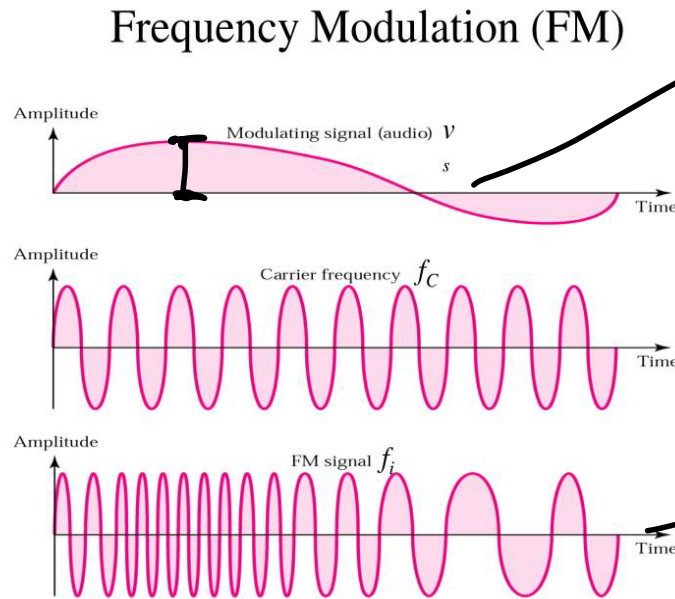


amplitude of signal varies

frequency changes over time



Frequency Modulation (FM)

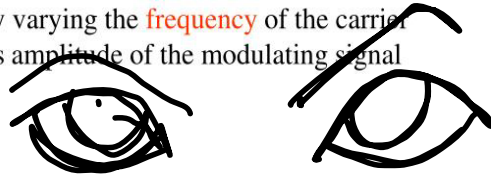


used to modulate the carrier wave.

In frequency modulation, the frequency of the carrier is varied following the instantaneous value of the amplitude of the modulating signal keeping the amplitude of the carrier fixed.

carrier signal after frequency modulation

In FM, the information is conveyed by varying the **frequency** of the carrier signal f_c in step with the instantaneous amplitude of the modulating signal V_s .

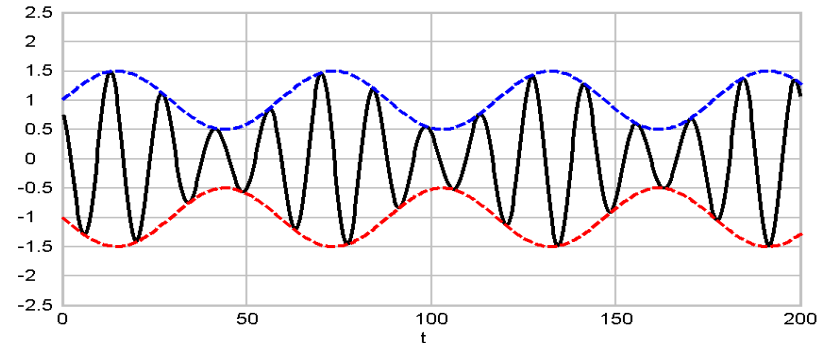
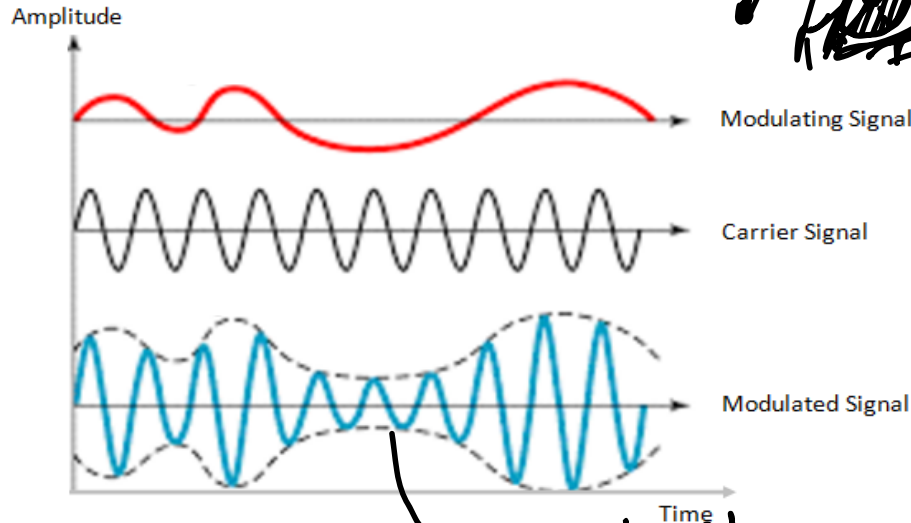
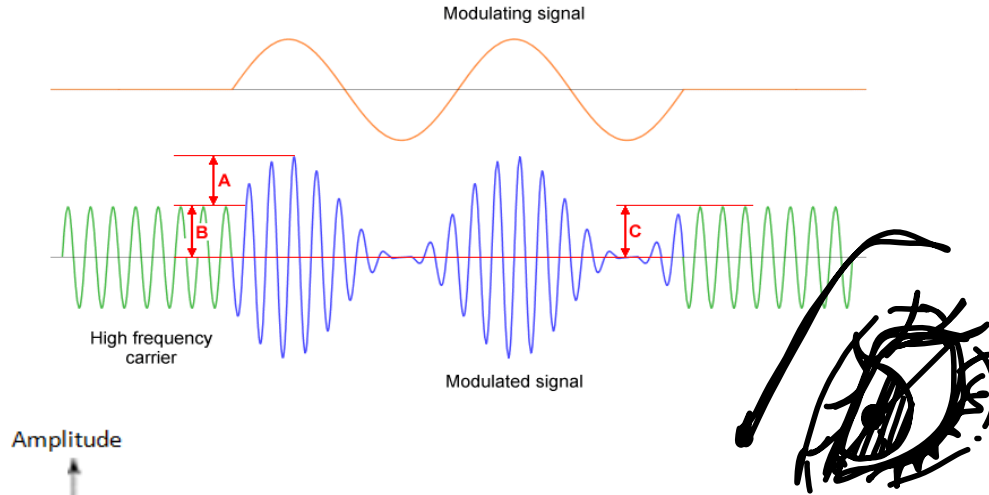


Modulation Depth in Amplitude Modulation (AM)

higher the depth, greater the change.

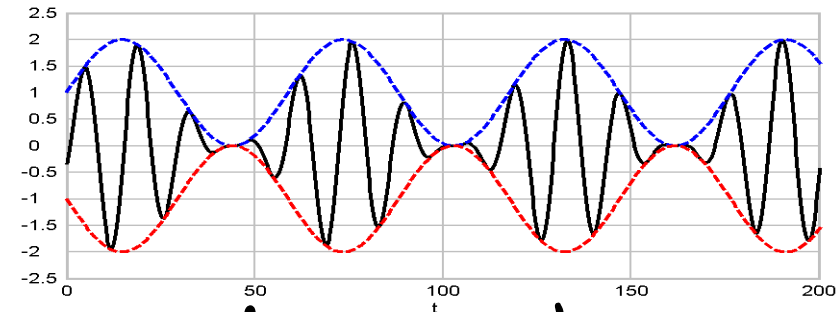
$$\text{Modulation Depth (\%)} = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \times 100$$

- Lower Modulation Depth ($1.5 - 0.5 / 1.5 + 0.5 = 0.5$ or 50%)
The amplitude of the carrier wave only changes **slightly**.



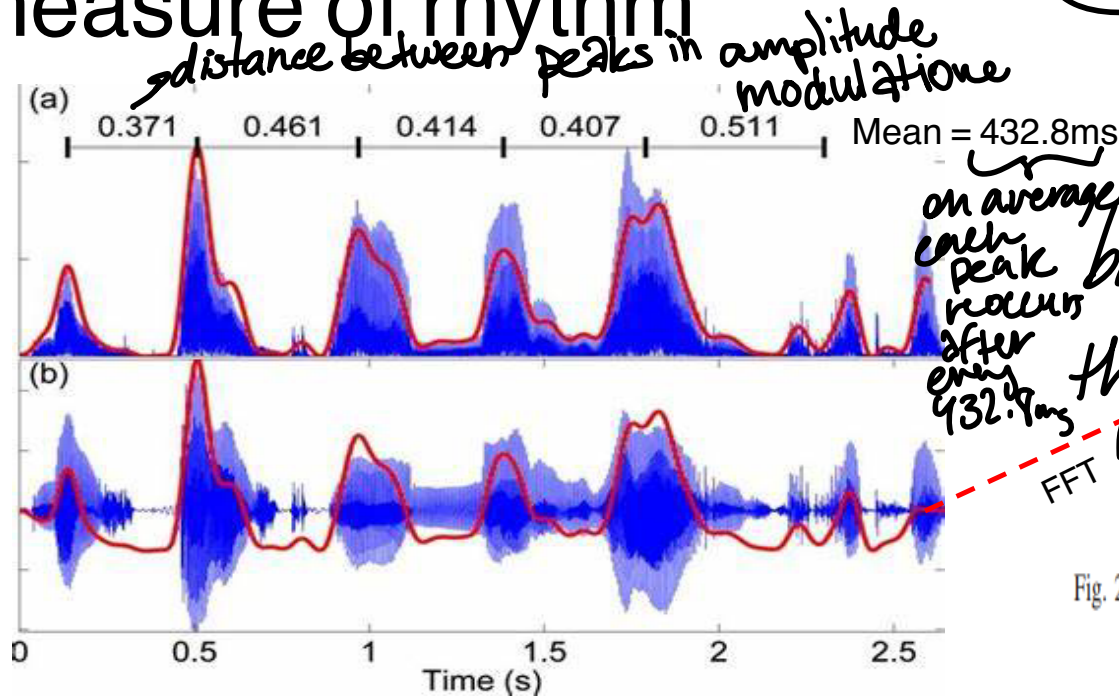
- Higher Modulation Depth ($2.0 - 0.0 / 2.0 + 0.0 = 2/2 = 1$ or 100%)

The amplitude of the carrier wave only changes **a lot**.



extract amplitude envelope and

Envelope Modulation Spectrum (EMS) as a measure of rhythm



“at least based on money raised it looks like...”

Fig. 1. (Color online) (a) Amplitude envelope superimposed over magnitude of bandpass-filtered signal; intervals between the several most prominent peaks in the amplitude envelope are shown. (b) windowed, mean-subtracted amplitude envelope over original acoustic signal.

From Tilsen and Johnson 2008

do the FFT analysis.

Fourier analysis will show the frequency modulation rate that will be the highest. Peak at 2.3 Hz (1/period: 1000/432.8)

on average, each peak occurs after every 432.8ms

basically to see the waviness pattern,

FFT

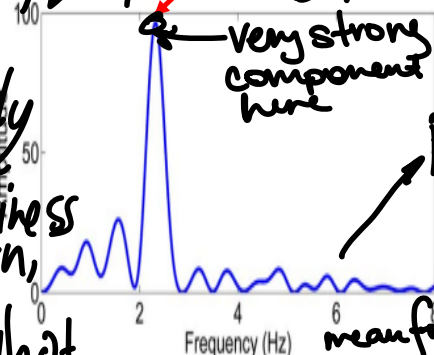


Fig. 2. (Color online) Power spectrum of the amplitude envelope that was shown in Fig. 1(b).

What are the frequencies of the prominent bumps?

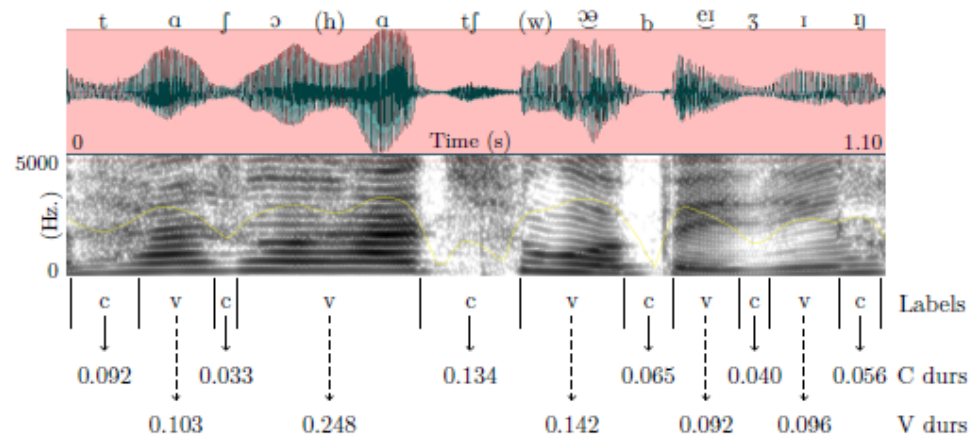
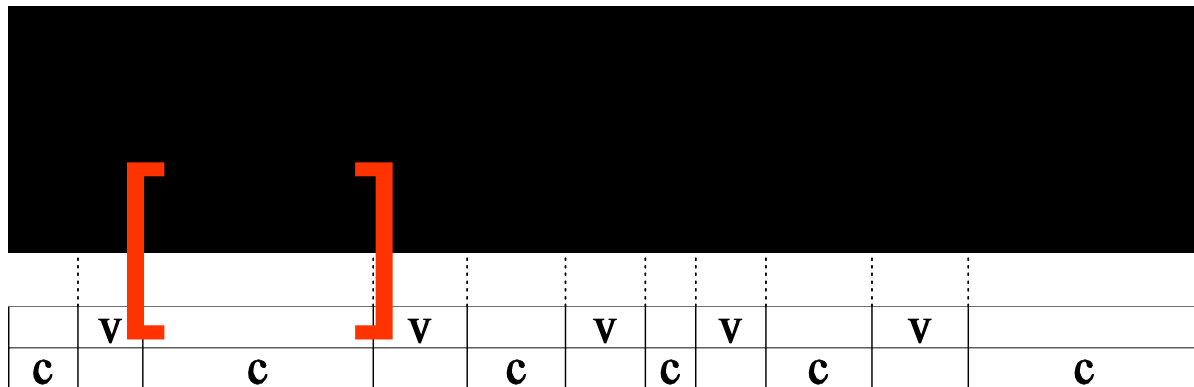
= 2.3 Hz.

corresponds to syllable rate (4-6 Hz)

*This is more like a foot
that occurs every 2.8 Hz.*

Traditional Linguistic Measures of Rhythm

- Long tradition of categorizing languages by rhythm class
 - Perceptual classification with problematic mapping to acoustic measures
- Proposal that these rhythm differences fall out of phonotactics or statistical regularities in strings of consonants and vowels
- Resulted in several sets of acoustic measures based on segmenting the speech stream into consonants and vowels



Measure Definitions

For the following definitions, X denotes a vector of consonantal or vocalic interval durations, while lowercase subscripted x_i represents the i^{th} duration value in the vector X . D is the vector containing all C and V durations for the utterance (summing yields the total speech duration). n gives the number of C or V intervals in X , while m represents the total number of C and V intervals in the utterance.

%, Ramus et al. (1999)

$$\%X = 100\% \times \frac{\sum_{i=1}^n x_i}{\sum_{j=1}^m d_j} \quad (1)$$

Δ , Ramus et al. (1999)

$$\Delta X = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(x_i - \left(\frac{\sum_{j=1}^n x_j}{n} \right) \right)^2} \quad (2)$$

rPVI, Grabe and Low (2002)

$$XrPVI = \frac{\sum_{i=1}^{n-1} |x_i - x_{i+1}|}{n - 1} \quad (3)$$

nPVI¹, Grabe and Low (2002)

$$XnPVI = 100 \times \frac{\sum_{i=1}^{n-1} \frac{|x_i - x_{i+1}|}{\left(\frac{x_i + x_{i+1}}{2} \right)}}{n - 1} \quad (4)$$

Varco, Dellwo (2006)

$$VarcoX = 100\% \times \frac{\Delta X}{\left(\frac{\sum_{i=1}^n x_i}{n} \right)} \quad (5)$$

%V: percentage of duration taken up by vocalic intervals; ΔC standard deviation of the duration of consonantal intervals within a sentence

F. Ramus et al. / Cognition 73 (1999) 265–292

273

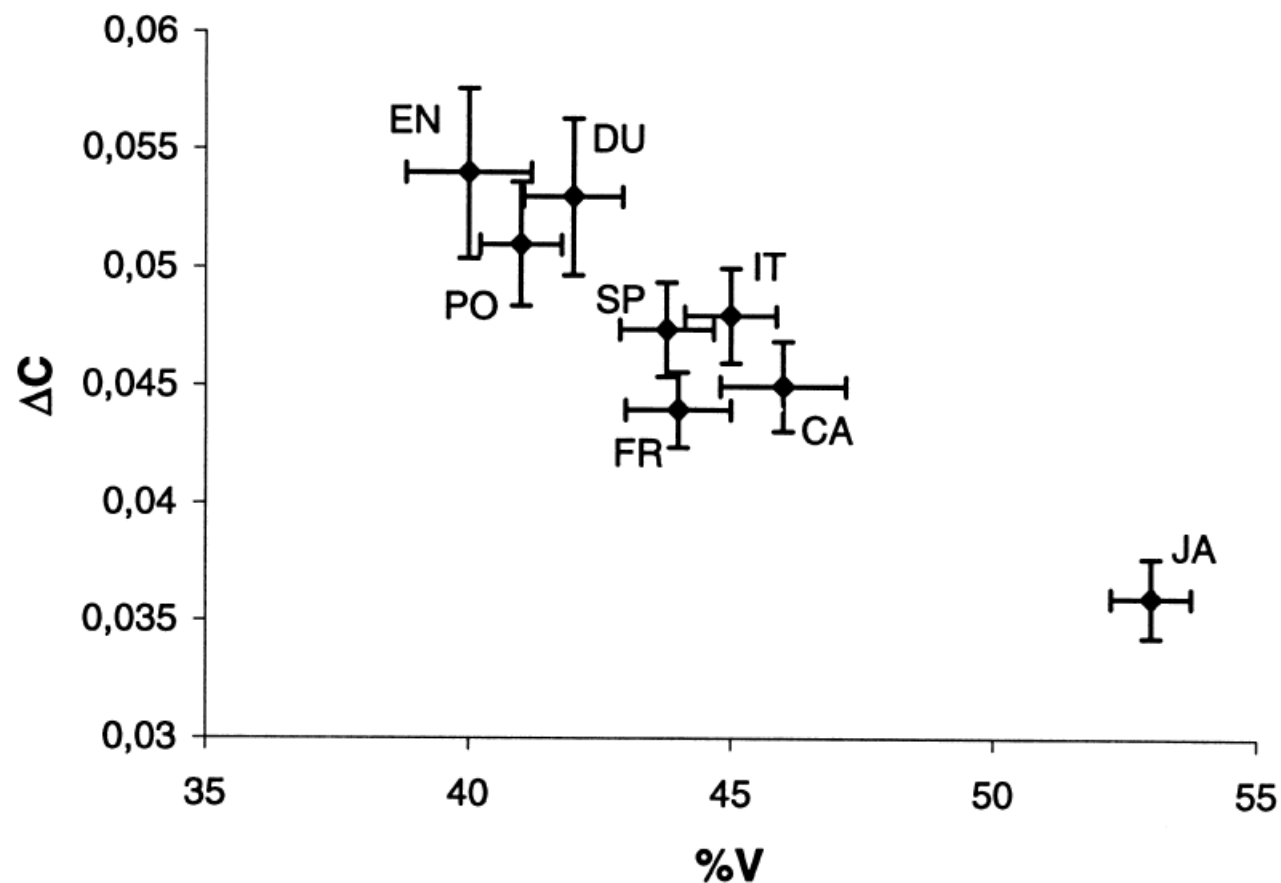
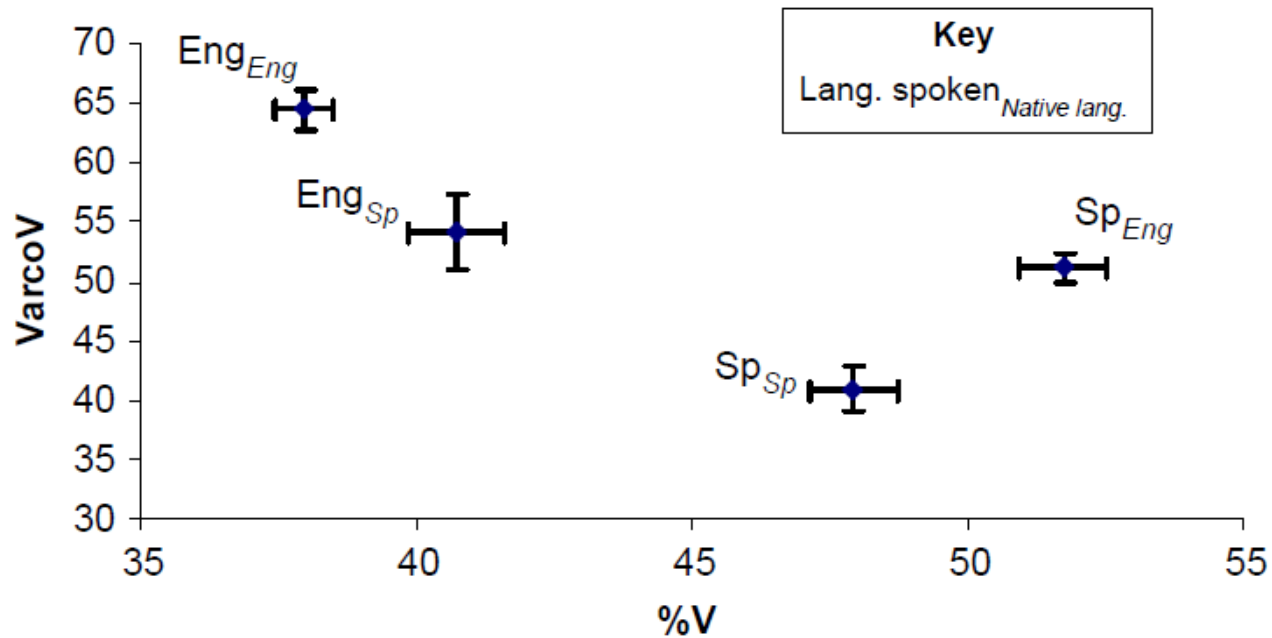


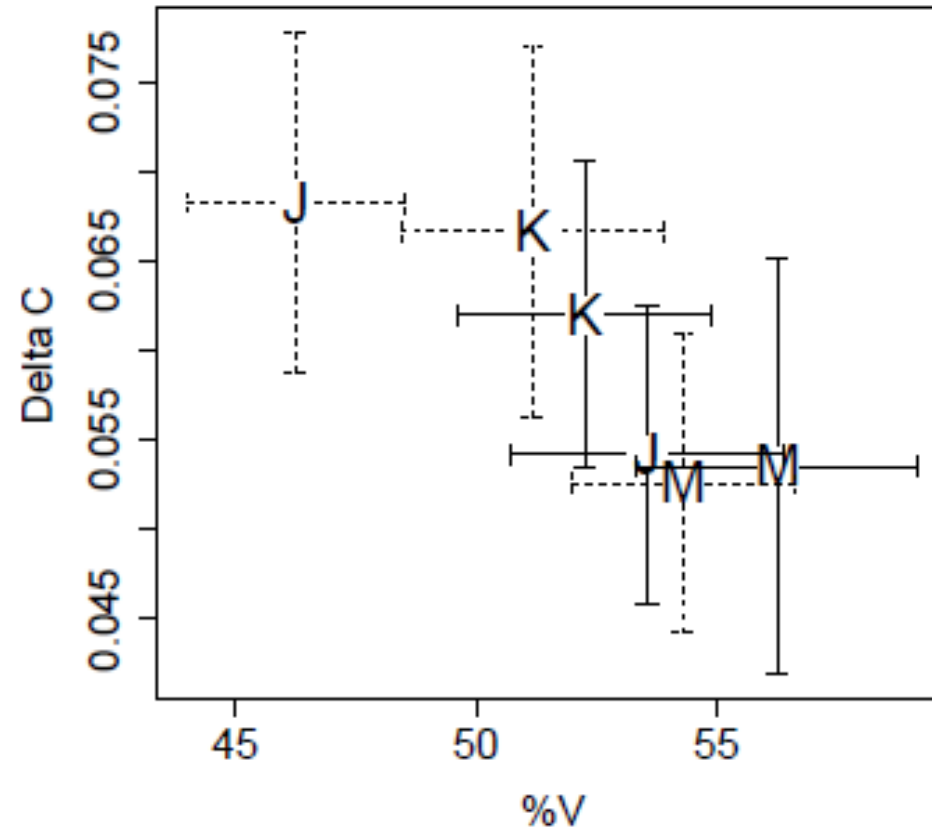
Fig. 1. Distribution of languages over the (%V, ΔC) plane. Error bars represent ± 1 standard error.

%V: percentage of duration taken up by vocalic intervals; VarcoV – Standard Deviation of Vowel Interval Durations divided by mean vowel duration

L. White, S.L. Mattys / Journal of Phonetics 35 (2007) 501–522



Consonantal Variability by Percent Vocalic (AC)



Japanese, Korean, Mandarin (Conversational and Reading) – Brenner (2016)

Problems with traditional measures of rhythm in speech

- Time consuming to measure (must mark consonant and vowel boundaries)
- Presumes that linguistic entities are the basis for the rhythm (ex.) foot, syllables, and word
→ may not be what we are actually perceiving.
- Difficulty dealing with non-linguistic sounds such as silent gaps or coughs → how do you deal with this?
- Difficult computing on non-clear or disordered speech

Principles of the Envelope Modulation Spectrum (EMS) “Approach”

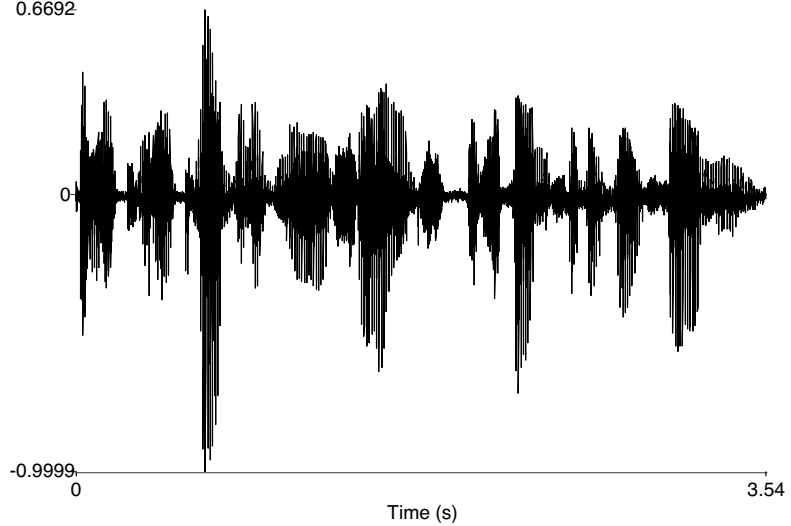
1. Rhythmic distinctions are not constrained solely by the structure of the language but also can be measured for individual differences within a language
2. The real determiner of rhythm is in the amplitude envelope of the speech signal
3. These rhythms are not just present in the overall amplitude envelope but within bands across the frequency spectrum
4. Rhythmic structure cannot be captured by a single number but depends on what is being classified. Rhythm is a complex percept which requires flexible measurement

Additional Benefits of the EMS approach

- Can accommodate non-linguistic sounds such as pauses, coughs, etc.
- Does not require one to make decisions on boundaries between vowels and consonants
- Is automated!!!! (Matlab program)
- Works with music as well as speech

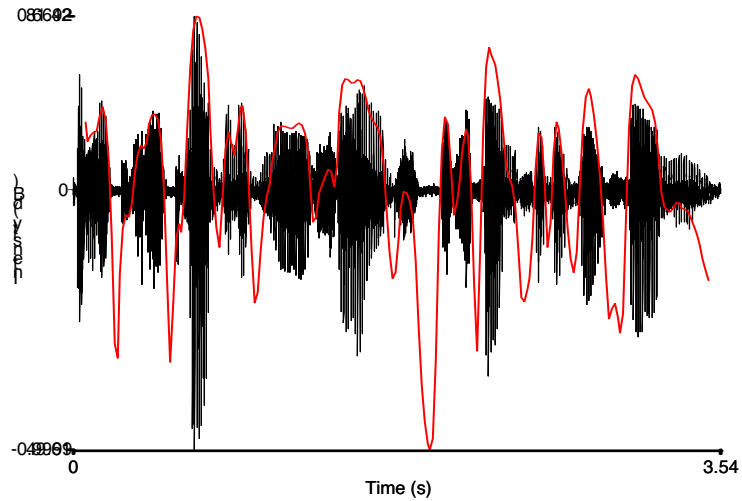
Maybe I want to use it.

0.6692

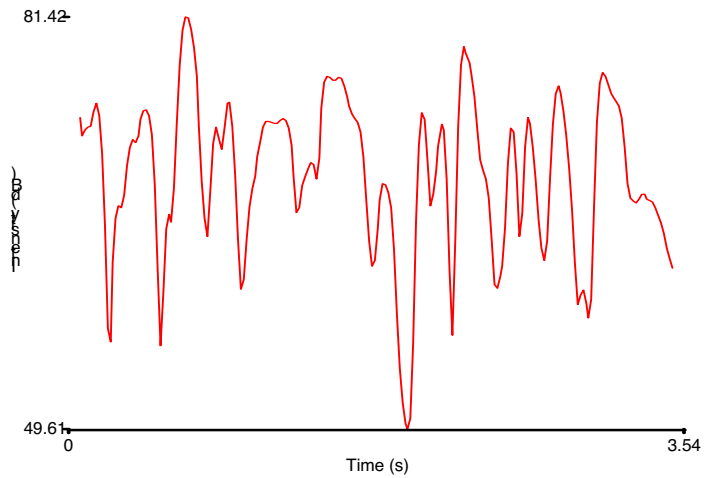


Amplitude Envelope

0.6692



81.42



0.6692

0.75

0

0

-0.9999

0



81.42

Pressure

49.61

0

Time (s)

3.54

88.812

Pressure

49.65

0

Time (s)

3.54

79.45

0

Time (s)

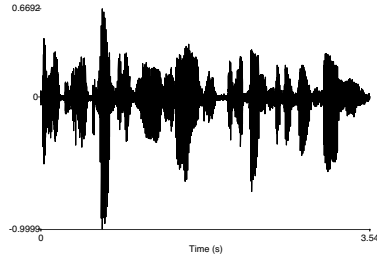
100 Hz



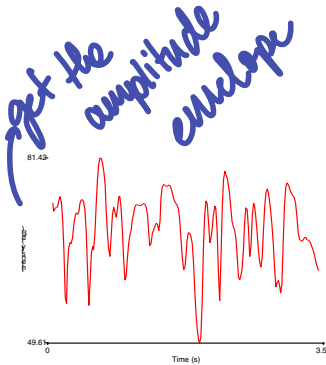
3.5

Envelope Modulation Spectrum

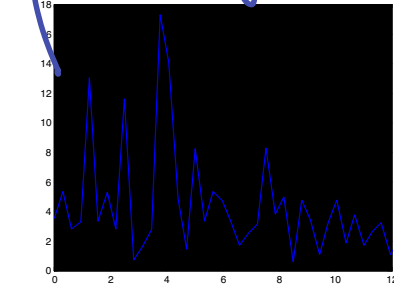
Envelope is a waveform – We can find the frequencies in it



air pressure variations



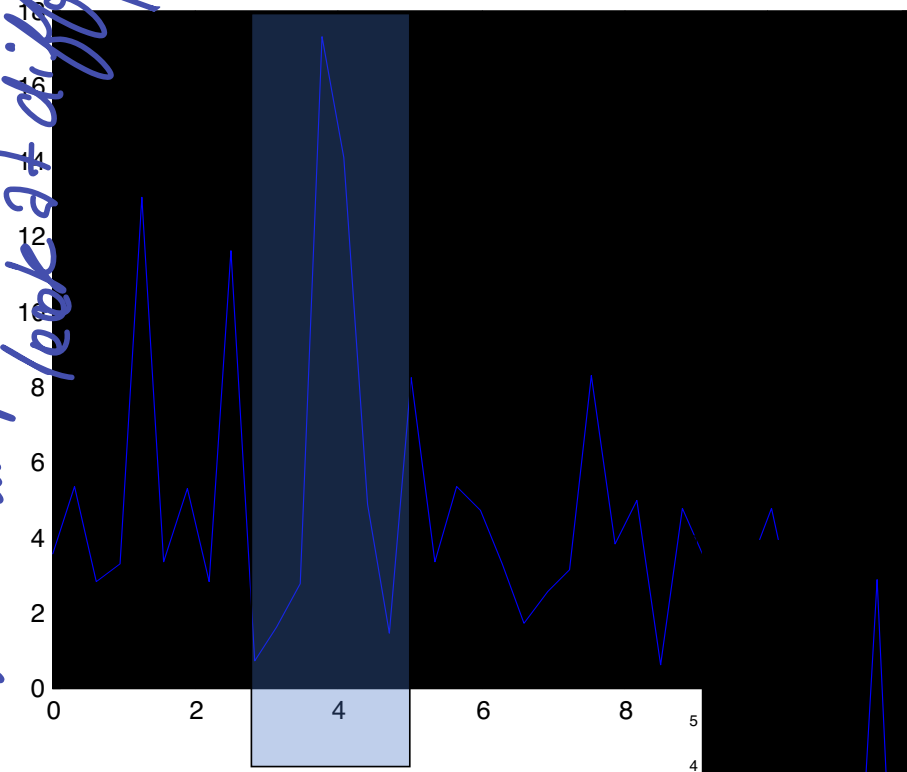
Extract Envelope



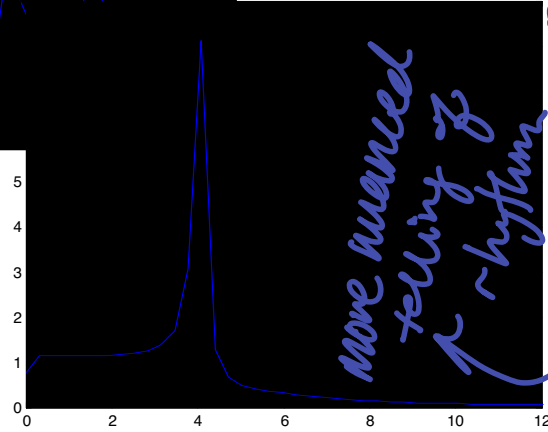
Calculate Spectrum
(Fourier Analysis)

do Fourier analysis on amplitude envelope.
Shows the peaks of the modulations.
Find highest amplitude peak by frequency.
 $\frac{1}{f_{avg}} = \text{period}$
around the time

we can extract the
 frequency component of the
 amplitude, we can
 look at different
 people



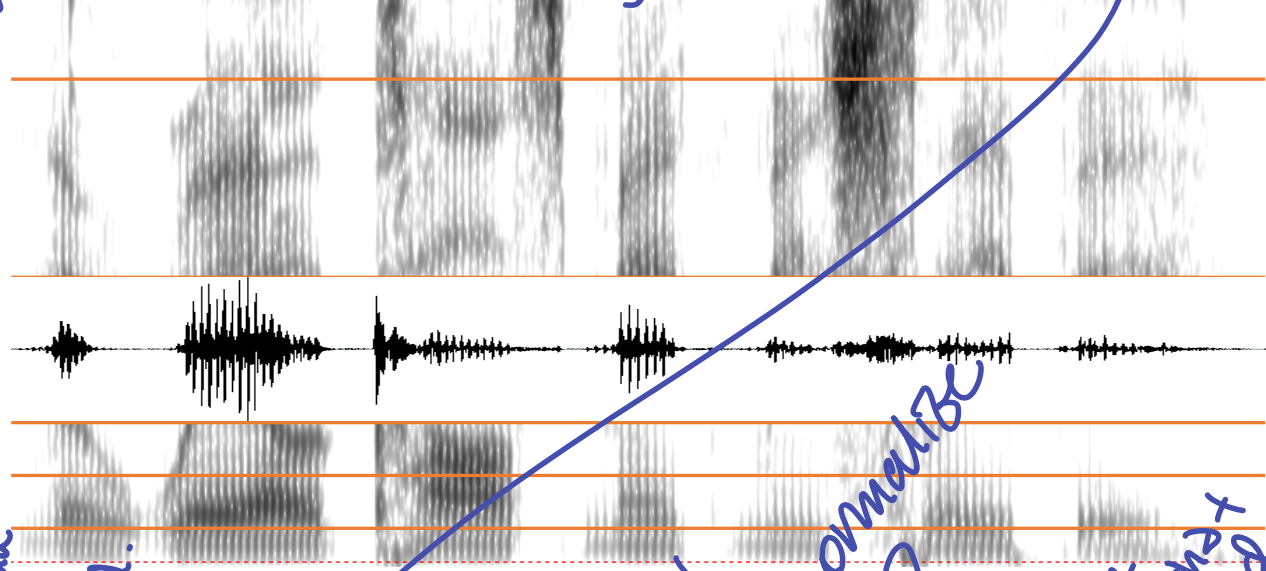
3-6hz
 14hz is
 around
 a syllable
 size
 we can infer that
 there is an amplitude
 modulation around
 every syllable
 size.



more nuanced
 telling of
 rhythm

you can customize the way you cut up the frequency bands, but your recovery should be supported.

Cut frequency into multiple bands



looking @ amplitude modulations & specific frequency bands.

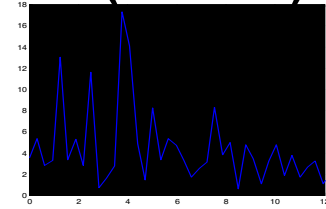
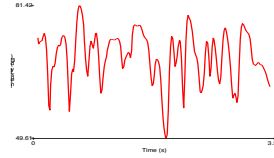
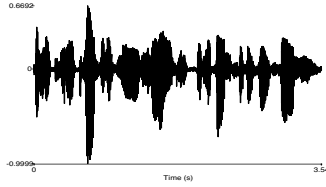
we don't really normalize for this?

look for what we do looking & multiple metrics?

The Envelope Modulation Spectrum (EMS)

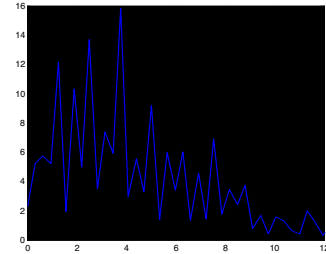
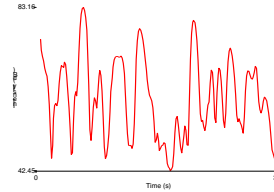
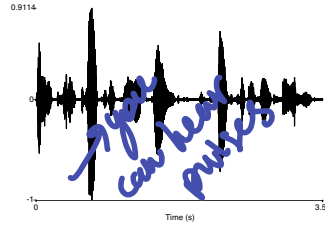
center of each band

Full Signal



Octave Bands

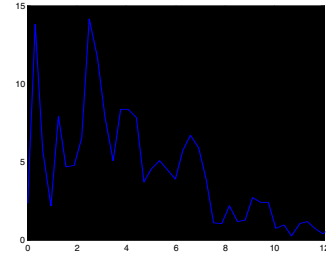
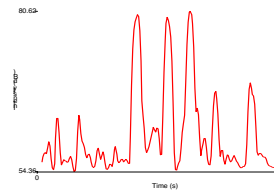
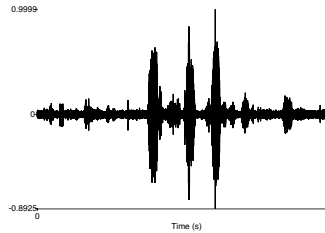
125



500

710-1420 Hz

1000

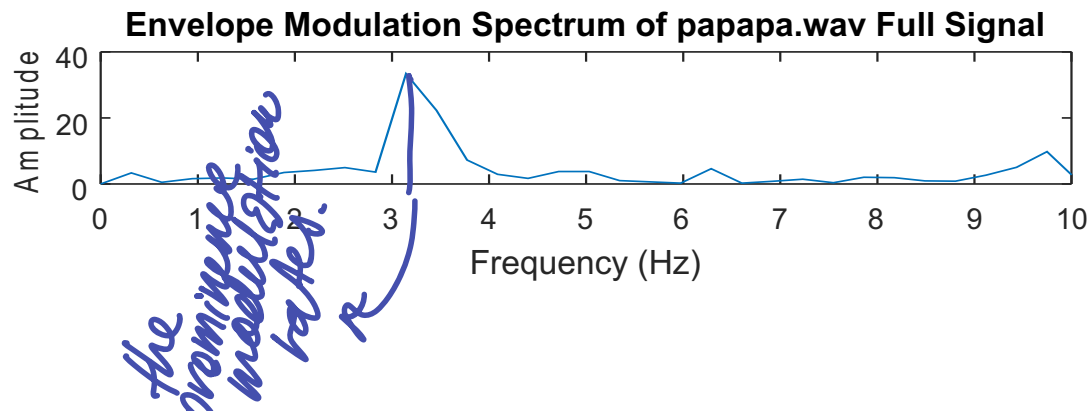
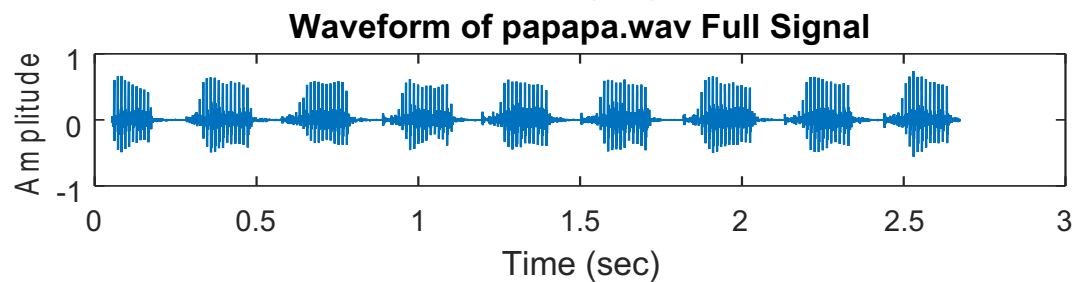
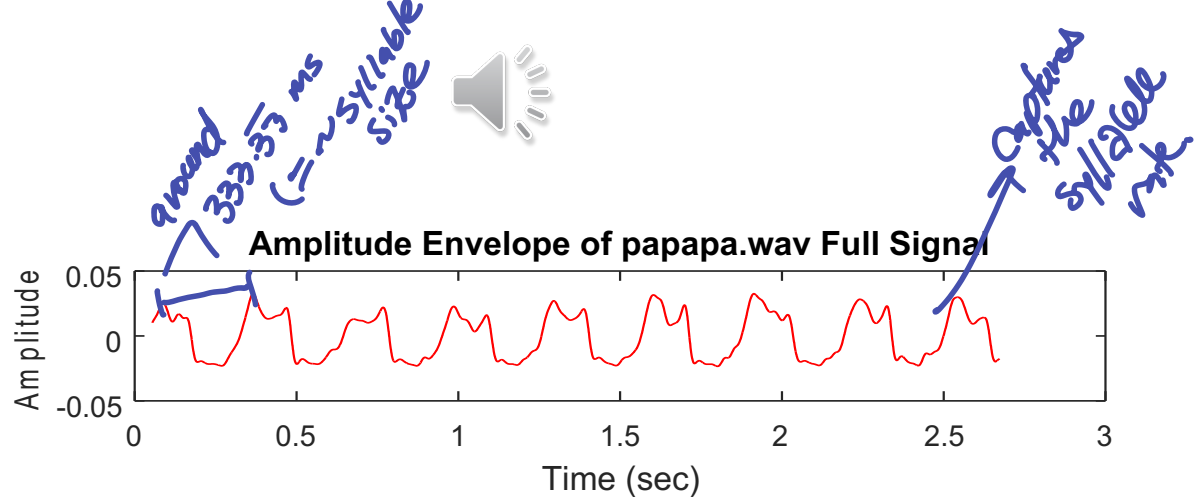


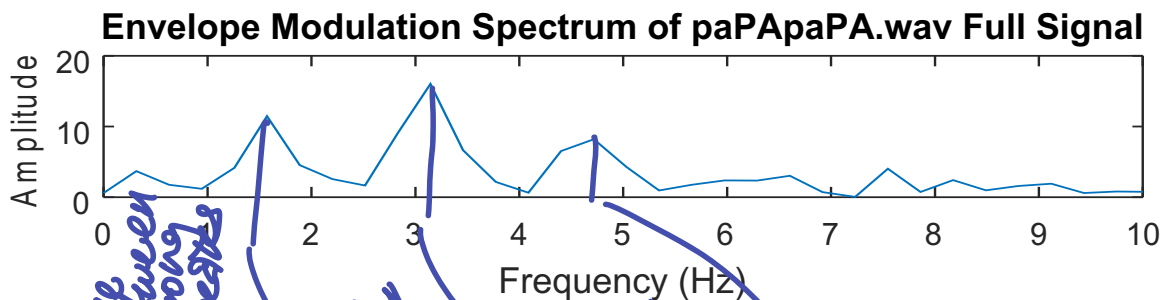
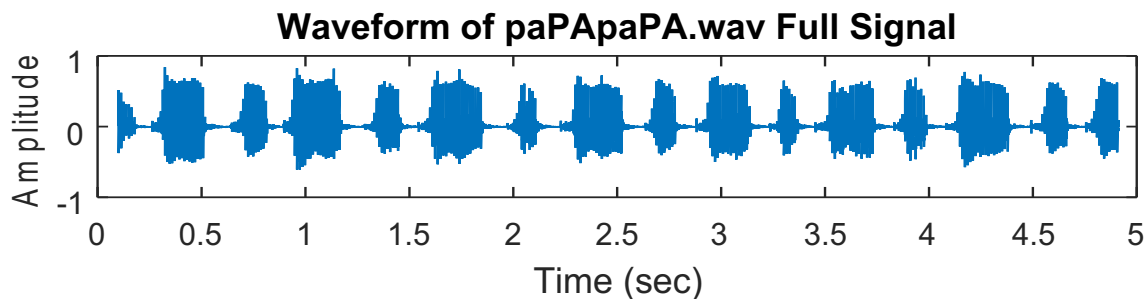
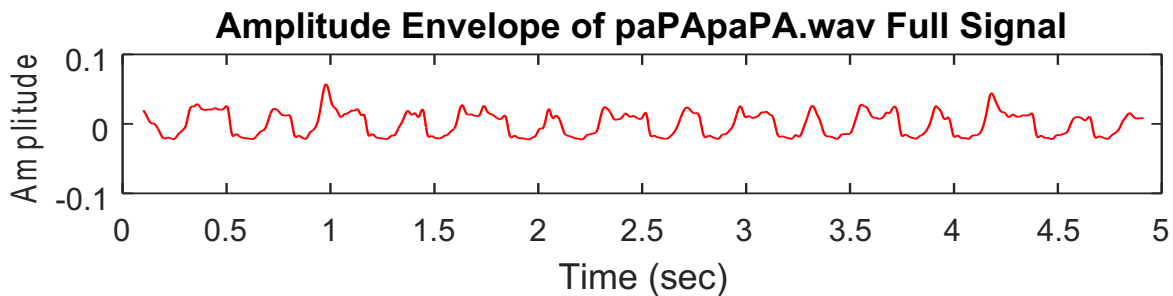
2000

4000

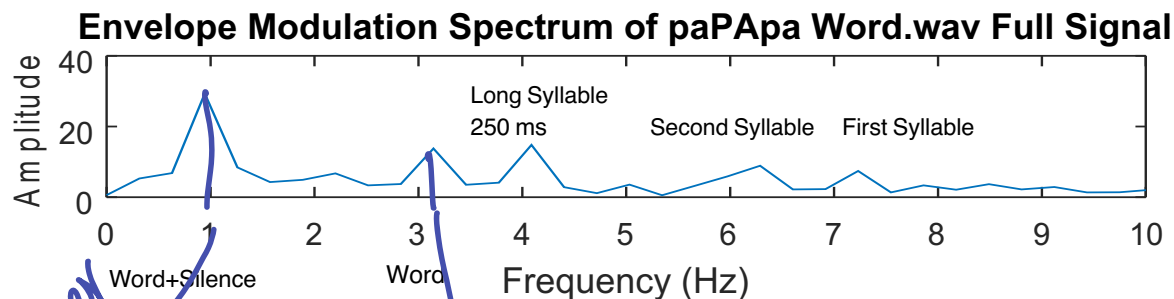
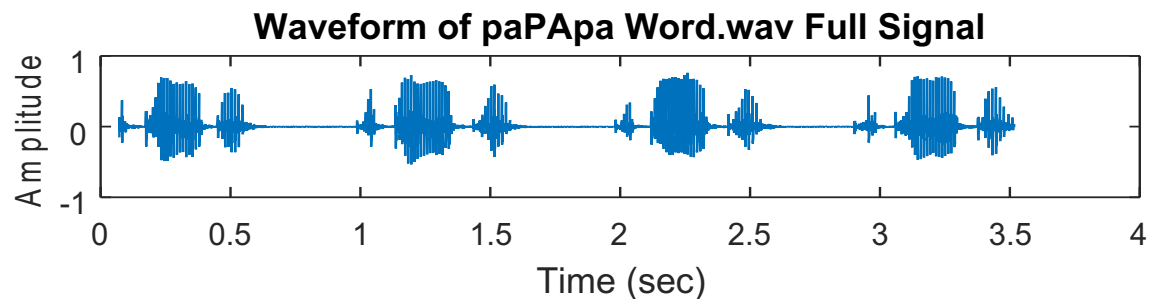
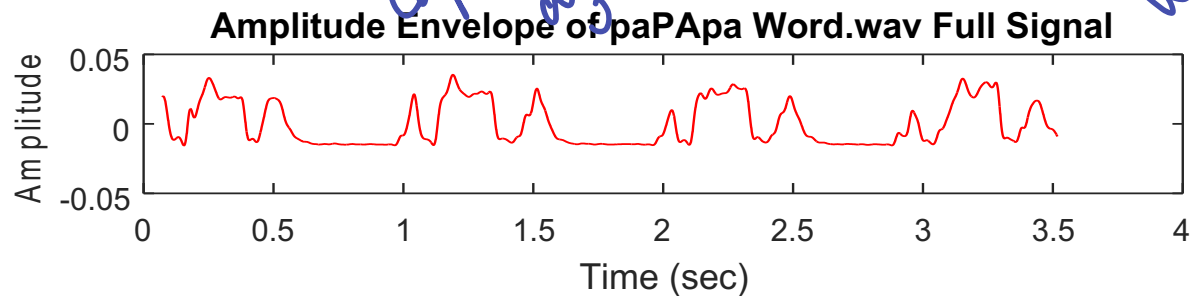
8000

5680-11,360 Hz





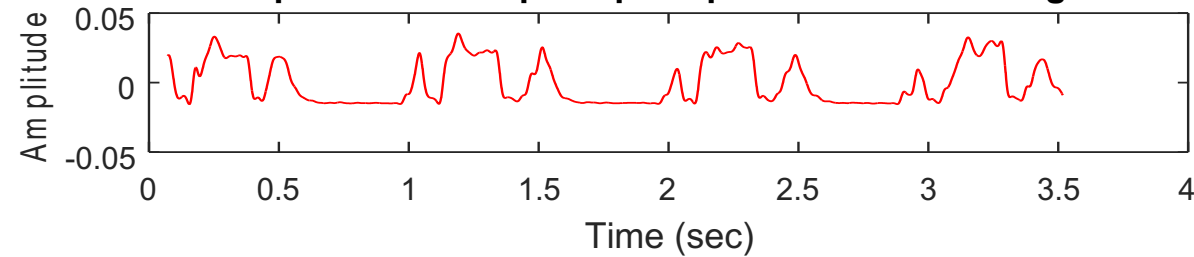
distance between strong peaks
but another rate still remains to be done



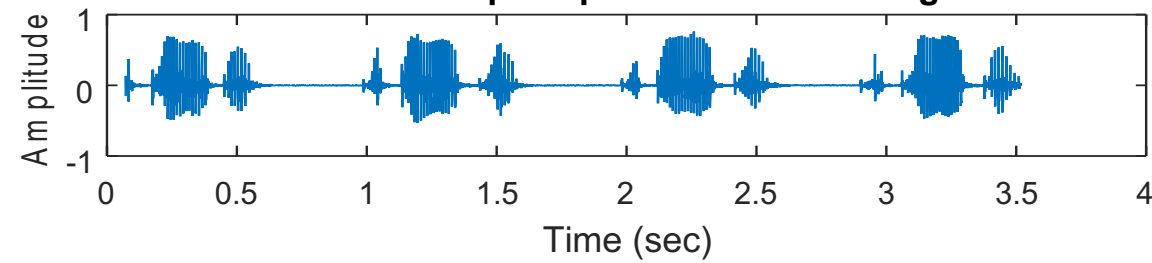
9
linguistic
like a unit,
foot or
a phrase
Syllable



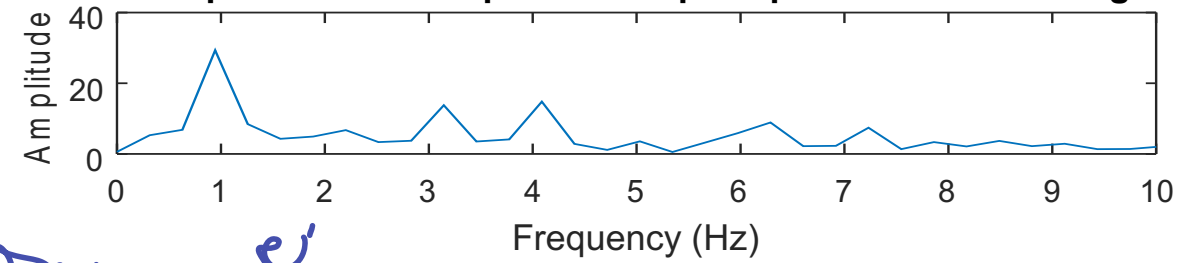
Amplitude Envelope of paPapa Word.wav Full Signal



Waveform of paPapa Word.wav Full Signal



Envelope Modulation Spectrum of paPapa Word.wav Full Signal

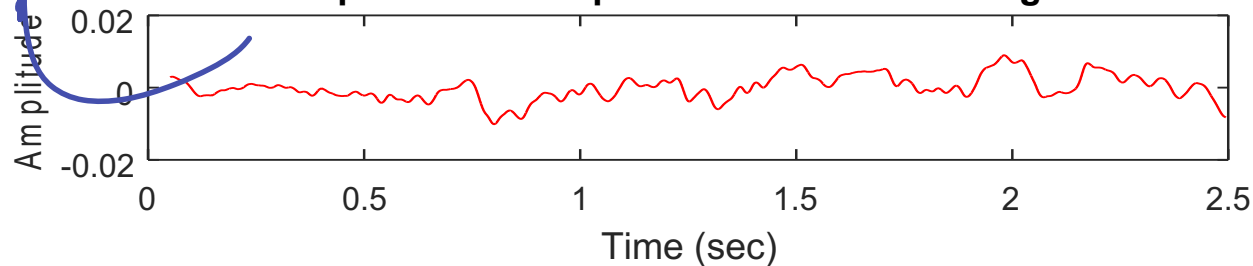


9 kHz
ms.
micro

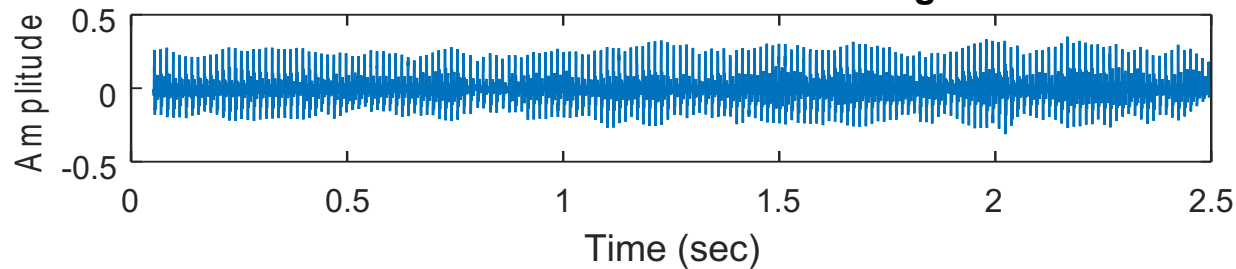
no repeated pattern
no rhythm



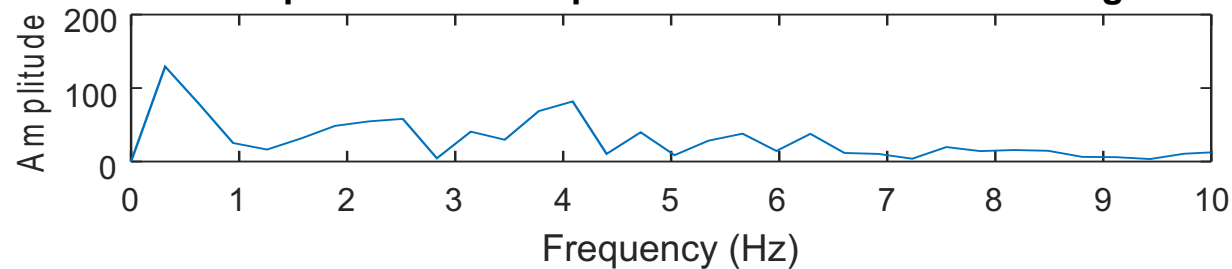
Amplitude Envelope of vowels.wav Full Signal

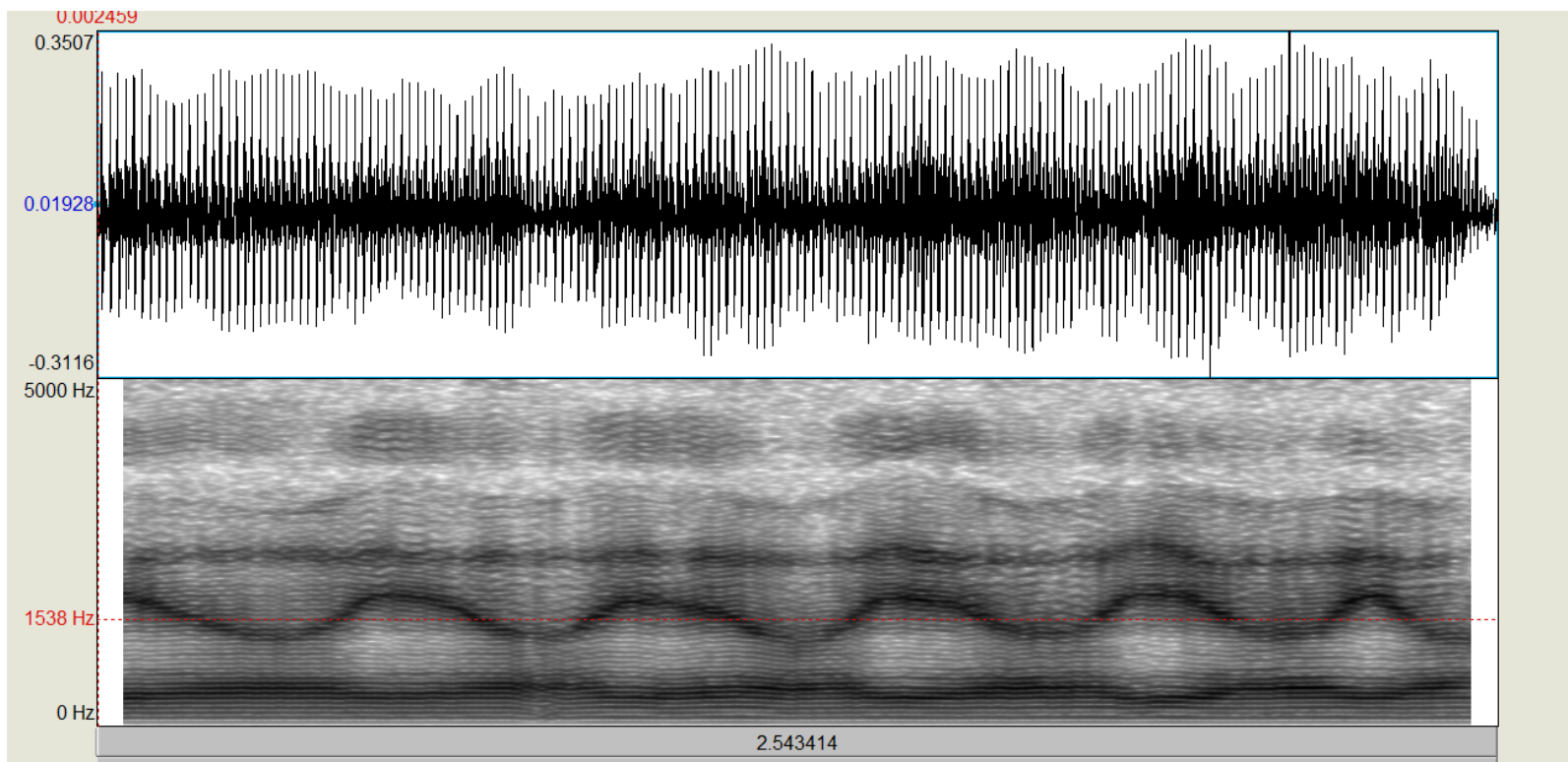


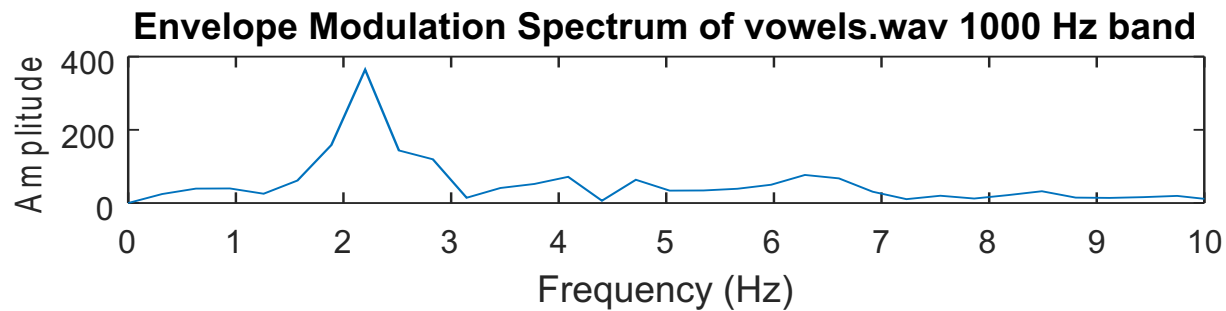
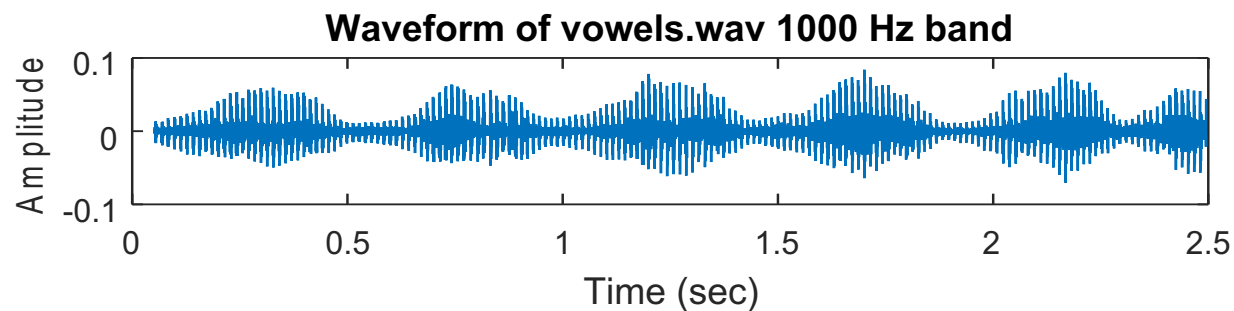
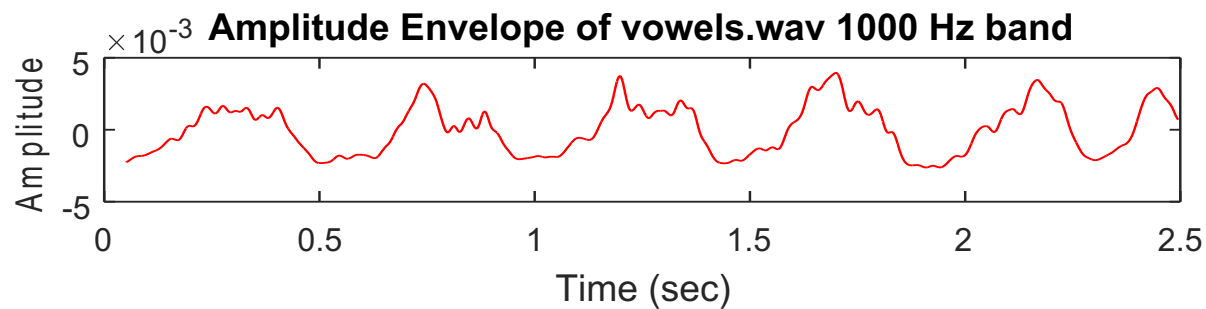
Waveform of vowels.wav Full Signal



Envelope Modulation Spectrum of vowels.wav Full Signal

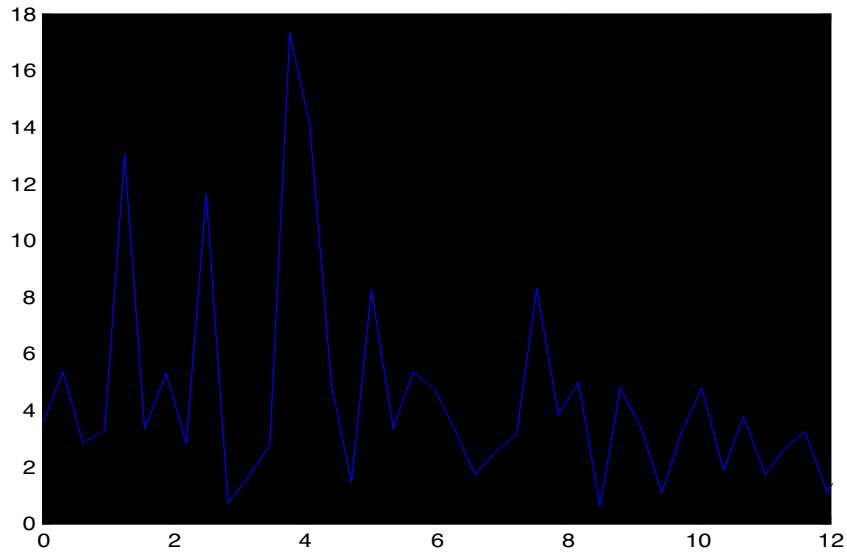






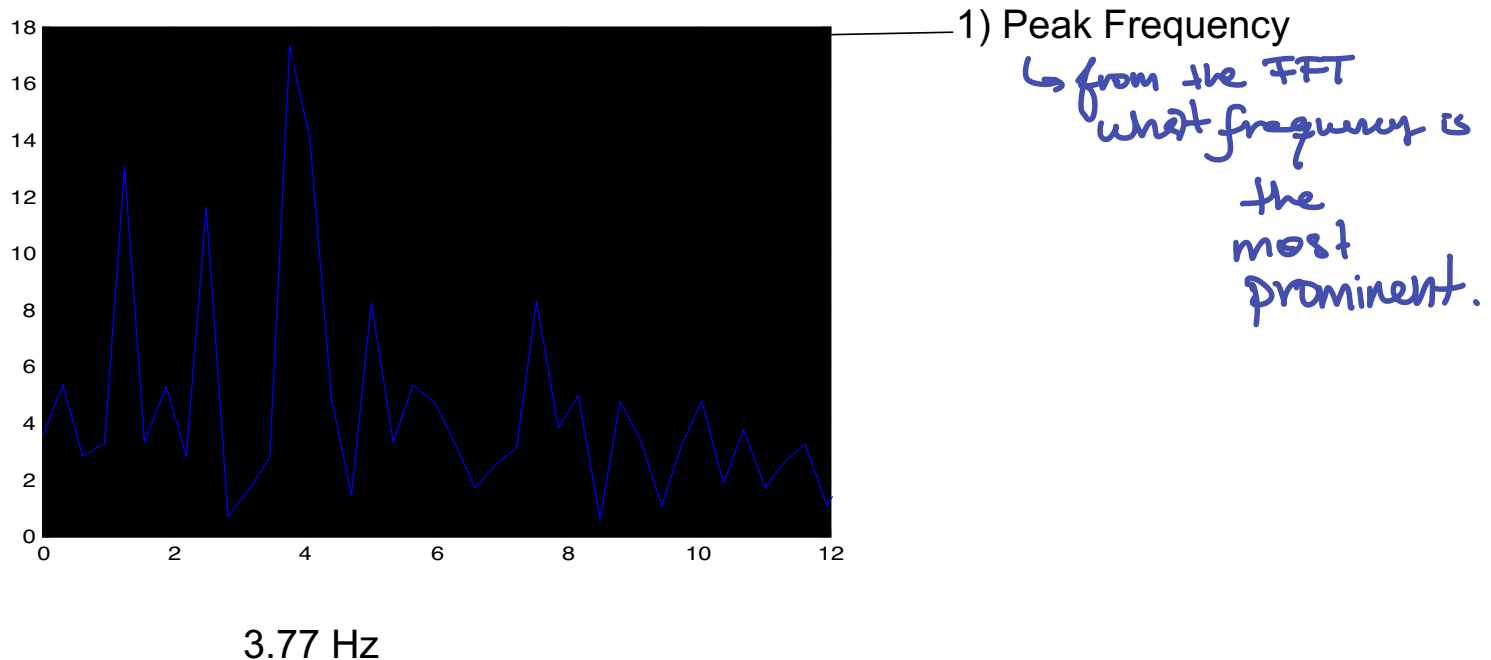
Dependent Variables

For full signal and each octave band, we calculate 6 dependent variables from the power spectrum:



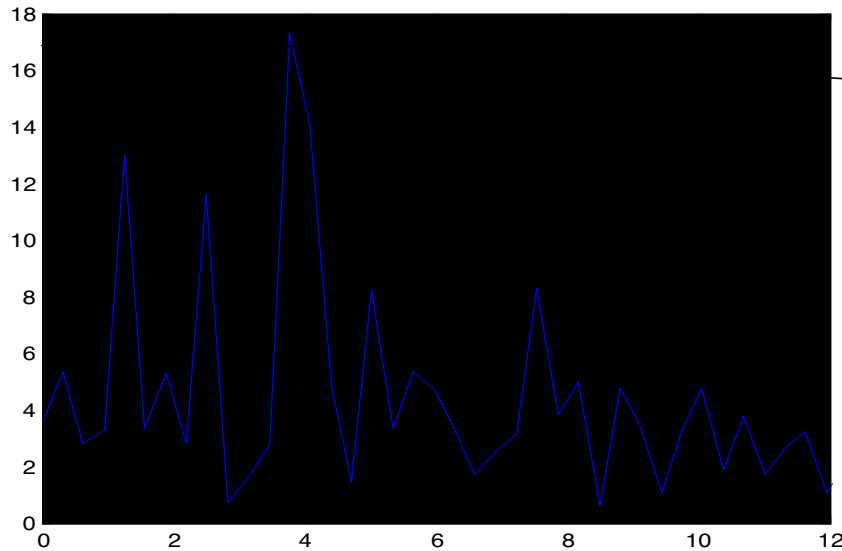
Dependent Variables

For full signal and each octave band, we calculate 6 dependent variables from the power spectrum:



Dependent Variables

For full signal and each octave band, we calculate 6 dependent variables from the power spectrum:



1) Peak Frequency

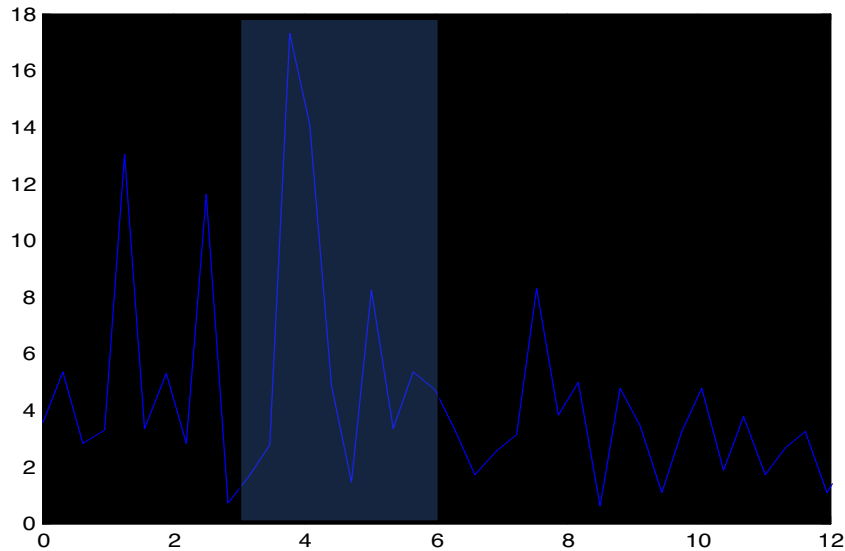
2) Peak Amplitude
(Normalized to Average amplitude)

↳ we want to
find how
consistent
it is

16.96 dB

Dependent Variables

For full signal and each octave band, we calculate 6 dependent variables from the power spectrum:

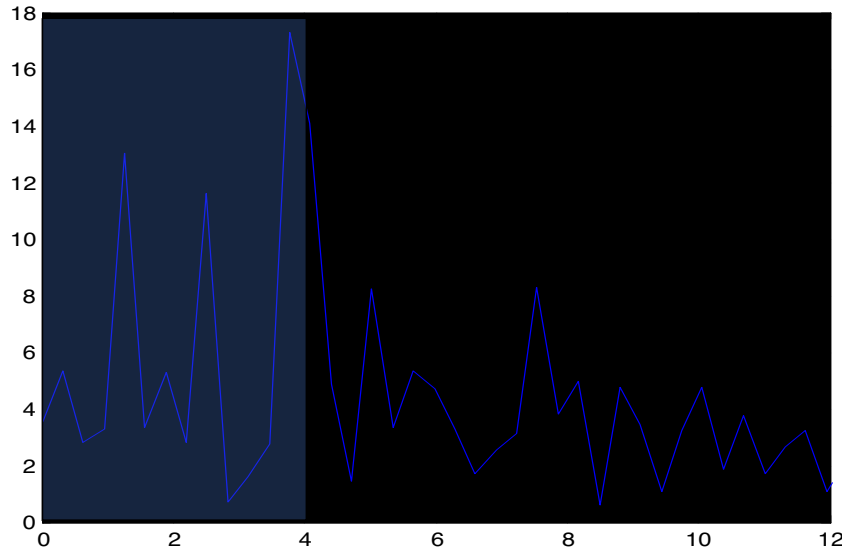


6.24 dB

- 1) Peak Frequency
- 2) Peak Amplitude
(Normalized to Average amplitude)
- 3) Average Energy Between
3-6Hz (Normalized)

Dependent Variables

For full signal and each octave band, we calculate 6 dependent variables from the power spectrum:

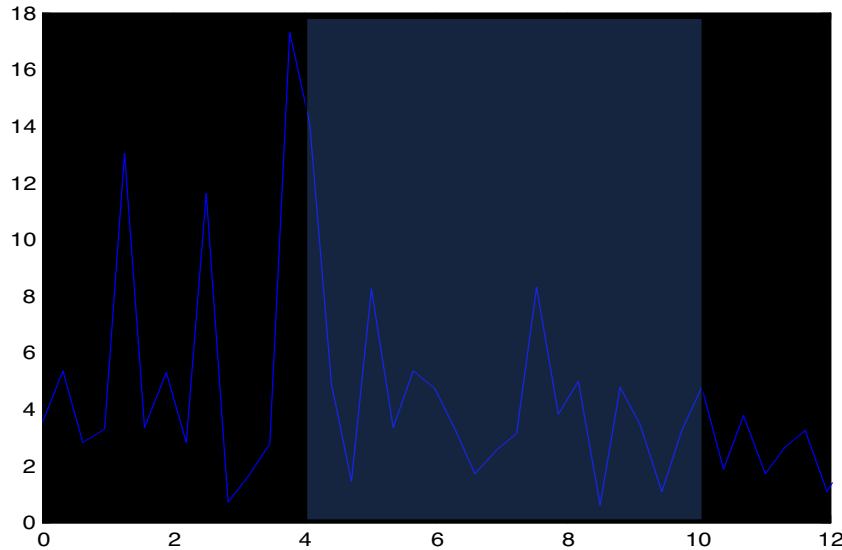


52.25 dB

- 1) Peak Frequency
- 2) Peak Amplitude
(Normalized to Average amplitude)
- 3) Average Energy Between
3-6Hz (Normalized)
- 4) Summed Normalized
Energy Below 4 Hz

Dependent Variables

For full signal and each octave band, we calculate 6 dependent variables from the power spectrum:

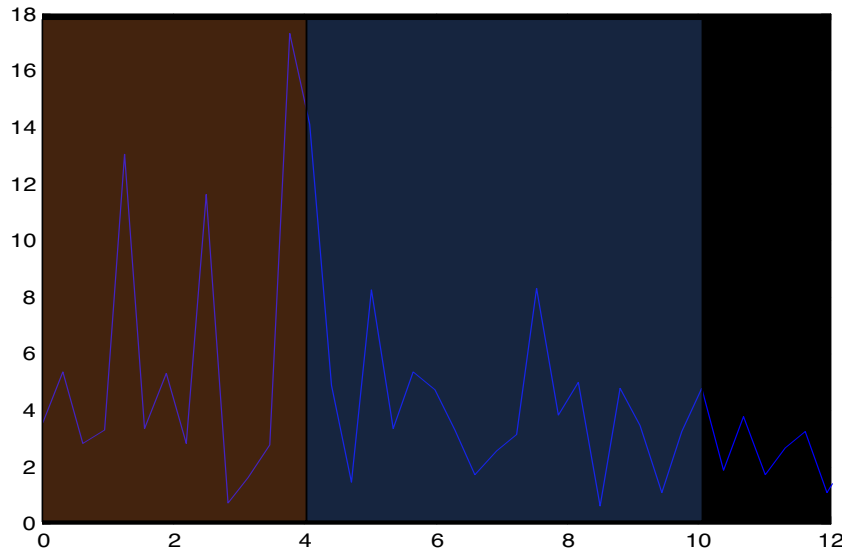


48.41 dB

- 1) Peak Frequency
- 2) Peak Amplitude
(Normalized to Average amplitude)
- 3) Average Energy Between
3-6Hz (Normalized)
- 4) Summed Normalized
Energy Below 4 Hz
- 5) Summed Normalized
Energy from 4 – 10 Hz

Dependent Variables

For full signal and each octave band, we calculate 6 dependent variables from the power spectrum:



1.08

- 1) Peak Frequency
- 2) Peak Amplitude
(Normalized to Average amplitude)
- 3) Average Energy Between
3-6Hz (Normalized)
- 4) Summed Normalized
Energy Below 4 Hz
- 5) Summed Normalized
Energy from 4 – 10 Hz
- 6) Ratio of Below/Above 4
Hz Energy

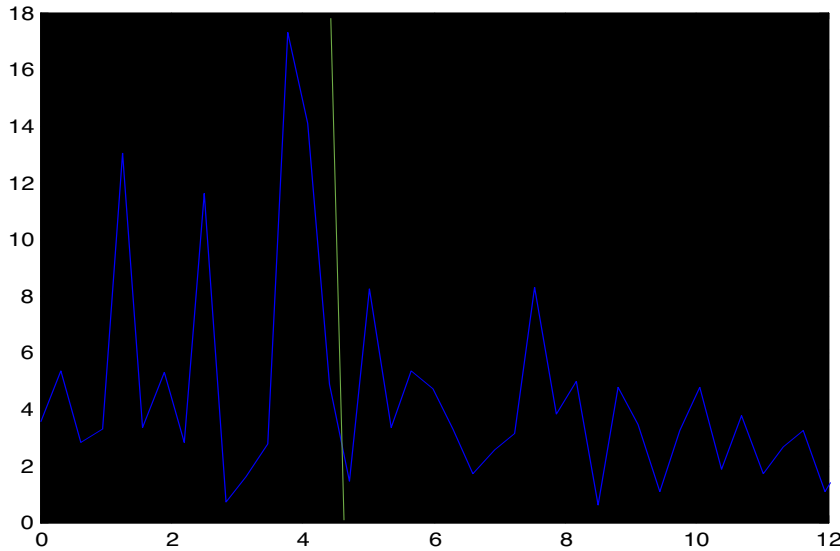
↳ to see how prominent
of the acoustic energy

maps on to
syllable
rate.

New Measure: Centroid

→ centre de gravité

The balance point of the spectrum if you take all of the frequencies and weight them by their amplitudes

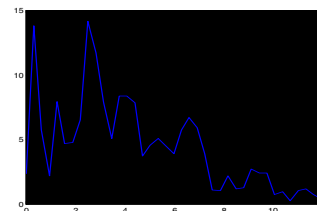
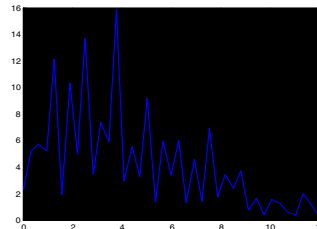
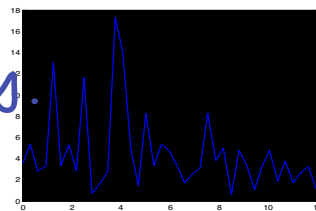


1.08

Across Bands

in Language Learners.

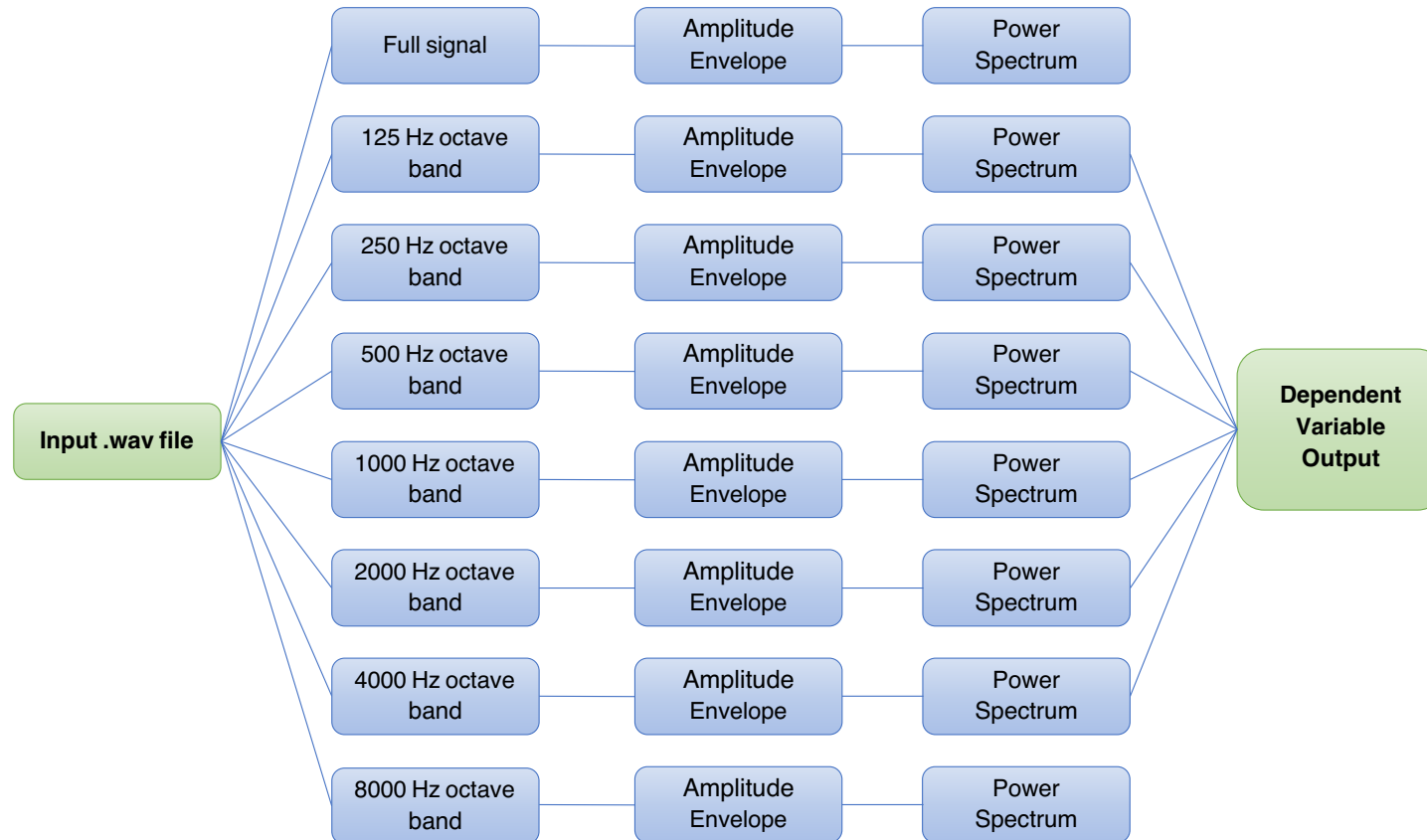
these can be the
dependent variables



	Full Signal	1000 Hz	8000 Hz
Peak Freq	3.77	3.77	2.52
Peak Amp	16.96	15.78	16.04
3-6 Hz	6.24	6.58	6.71
Below 4 Hz	52.25	66.40	67.08
Above 4 Hz	48.41	44.04	53.98
Ratio	1.08	1.51	1.24

NEED
TO TRY
THIS

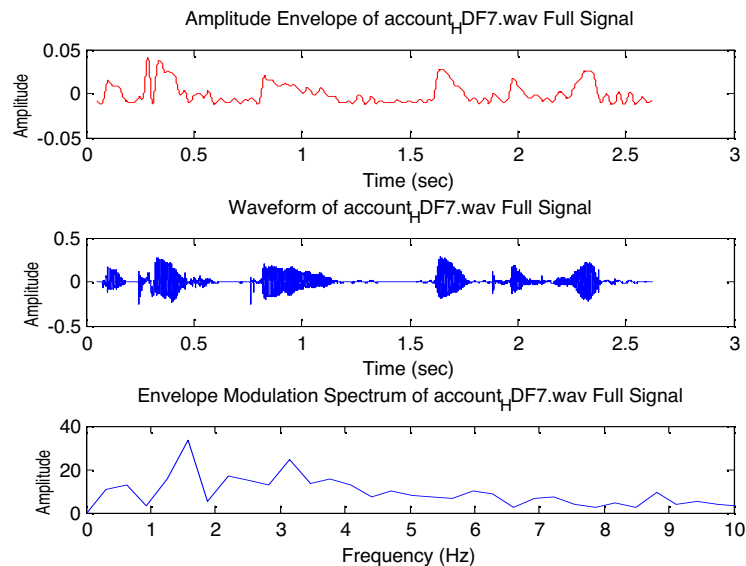
Overview of EMS



The EMS program can take in wav files in a batch from a folder and within seconds outputs into Excel

	Cent_All	Peak_Freq	Peak_Amp	E3_6Hz_all	Below40_a	Above40_a	Ratio40_all	Cent_125	Peak_Freq	Peak_Amp	E3_6Hz_12	Below40_1	Above40_1	Ratio40_12	Cent_250	Peak_Freq
Phrase1.	3.425827	0.317907	15.0091	6.541807	9.375663	2.910128	3.221735	4.547705	2.861159	9.275128	5.615776	5.203839	3.614044	1.439894	4.449759	2.543253
Phrase2.	4.335746	1.271626	12.82017	5.306656	6.889073	4.708672	1.463061	3.97203	1.589533	12.58165	5.406253	7.345126	4.397401	1.670334	4.569622	1.589533
Phrase3.	4.774308	5.722318	12.16901	6.227234	5.306768	5.394242	0.983784	3.687382	0.635813	6.879099	3.057573	4.133846	1.879141	2.199859	4.349054	2.225346
Phrase4.	4.338622	4.132785	12.61654	8.302274	7.582878	5.25892	1.441908	3.786828	0.317907	6.612182	2.97934	4.063382	1.990412	2.041477	3.979737	0.317907
Phrase5.	3.72628	2.524689	9.86315	5.735052	6.841426	3.076842	2.223522	4.252999	2.209103	11.07864	7.182079	6.819213	4.398353	1.550401	4.126455	2.524689
Phrase6.	4.786975	2.840275	7.667593	3.763845	4.318788	3.631681	1.189198	3.769411	2.840275	18.63536	5.456082	8.157107	3.600931	2.265277	3.958119	1.262344

Also can output figures for any band

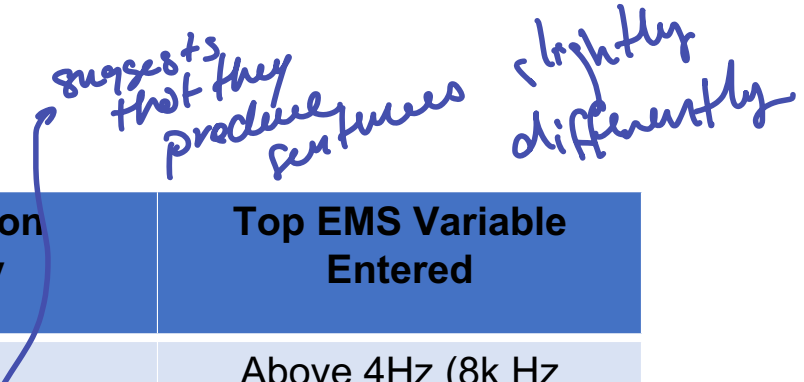


Experiments using EMS: English –Spanish Bilinguals

Methodology

- **Speakers:** 12 bilingual speakers (5 males; 7 females) of English & Spanish
- **Materials:** Each speaker produced the same 10 sentences; (5 English & 5 Spanish). Sentences taken from White & Mattys (2007)
- **EMS Analysis:** Each sentence from each speaker was analyzed with EMS program resulting in 48 variables for each sentence
- **Data Analysis:** Discriminant Function Analysis (DFA) was used to attempt to classify the sentences according to speaker using only the EMS variables.

Language Results



Analysis	Classification Accuracy	Top EMS Variable Entered
Individual Sentences (120 sentences)	78.3%	Above 4Hz (8k Hz Octave Band)
Means Across Talkers (24 Tokens – Averages calculated across 5 sentences for each talker -language)	95.8%	Above 4Hz (8k Hz Octave Band)

Chance = 50%

This variable is highly correlated with several linguistic measures

Talker Results

WAY above 8.3%

Analysis	Classification Accuracy	Top EMS Variable Entered
Talker recognition for Eng sentences	86.7%	Below 4k Hz (all octave bands)
Talker recognition for Span sentences	86.7%	Energy between 3-6Hz (8k Hz octave band)
Talker recognition across both languages	71.5%	Below 4k Hz (all octave bands)
Classifying talkers with means from both languages	83.3%	Peak Amplitude (250Hz octave band)

Chance = 8.3%

→ guessing

→ that it says which language it is.

→ and use to predict

→ they switch

rhythm in each lang differently.

↓
but still way above chance

The Acquisition of similar versus different speech rhythmic class (Wayland et al. In Press “Rhythms of Speech and Language: Culture, Cognition, and the Brain”)

Research Questions:

Do speakers of **rhythmically similar L1s** acquire L2 rhythm more easily?

The Acquisition of similar versus different speech rhythmic class

(Wayland et al. In Press “Rhythms of Speech and Language: Culture, Cognition, and the Brain”)

Methods – Stimuli

Data from **The BonnTempo Corpus**

L1: **German, English, French** (read speech at various rates)

L2 learners speaking each other's languages

(ex.) german speakers speaking Eng, etc...

they all say the same thing.

The Acquisition of similar versus different speech rhythmic class (Wayland et al. In Press “Rhythms of Speech and Language: Culture, Cognition, and the Brain”)

Results – L1 Rhythm Influence *only on EMS parameters*

L1 rhythm persists in L2:

German-accented English was rhythmically closer to native English than French-accented English. *→ b/c they share the rhythmic patterning.*

English-accented German resembled native German more than French-accented German.

Different rates affect rhythmic adaptation strategies.

→ they adapt the rhythm, but they

are still faithful to their
rhythm pattern.

Spectral variability: Long Term Average Spectrum (LTAS)

↳ measures intensity from
acoustic signal

Spectral Variability Measures

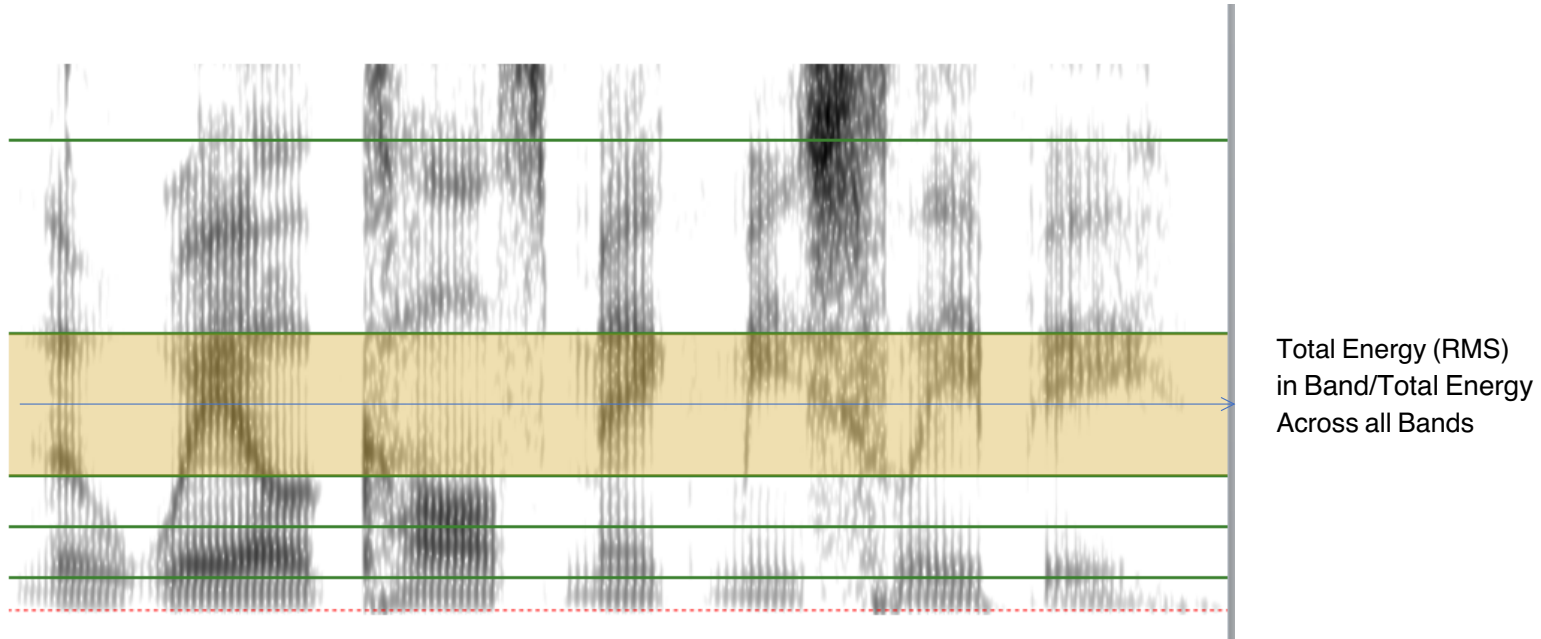
We have also developed a set of automatically calculated measures of spectral variability (named LTAS for Long Term Average Spectrum)

↳ we look @ Spectral energy regularity.

LTAS measures are similar to EMS in that they are computed for octave frequency bands. They are meant to provide measures of variability in the spectral energy (as opposed to EMS which measures temporal regularities)

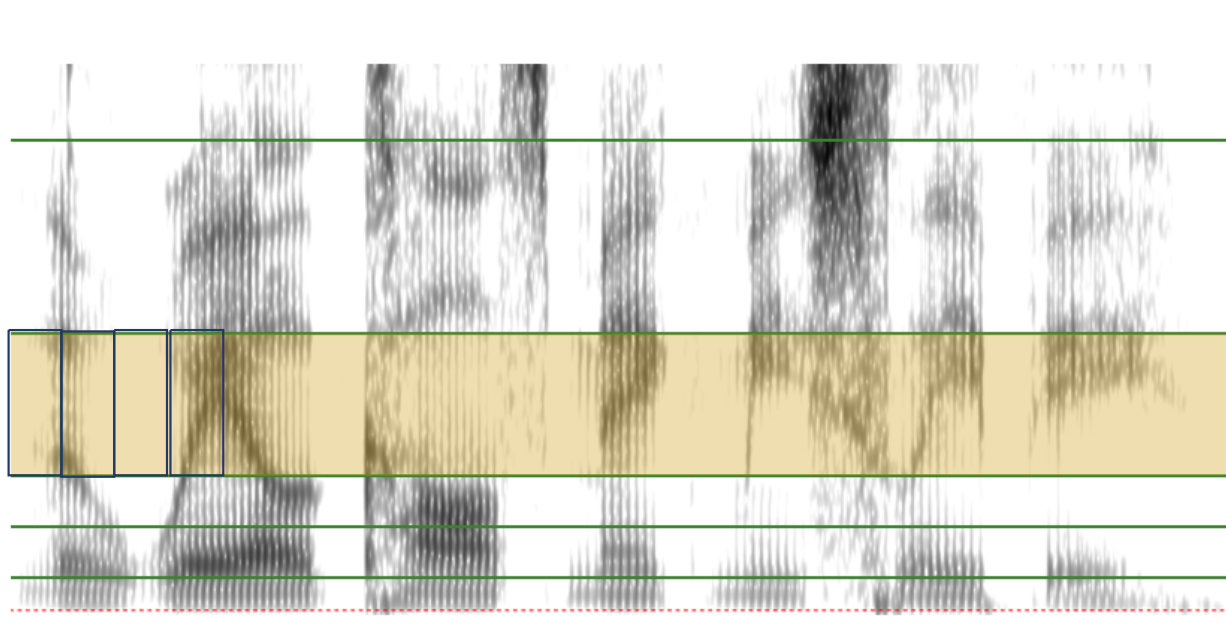
→ we want to quantify how acoustic energy changes over time.

LTAS Measures #1: Relative Energy in each Frequency Band



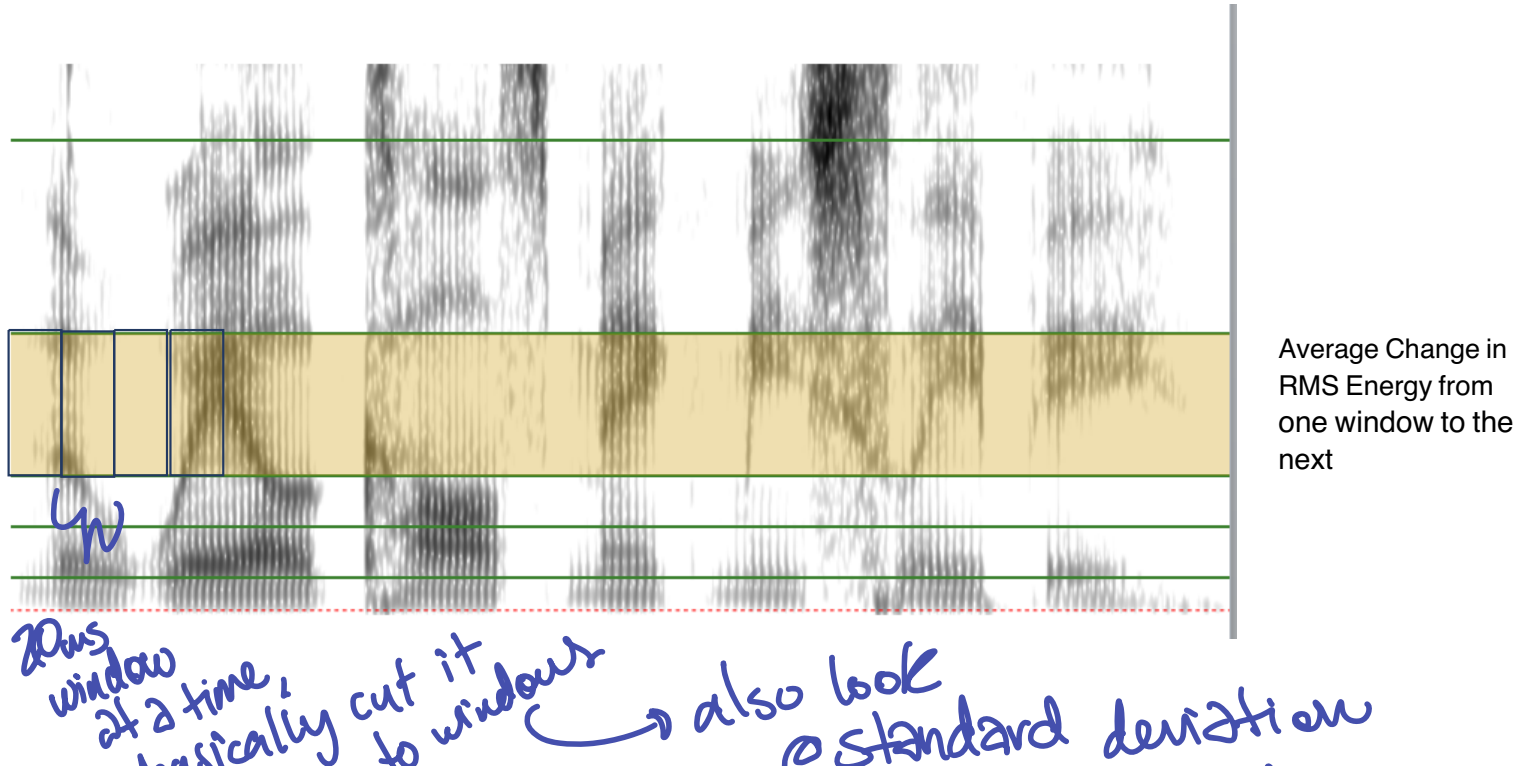
LTAS Measures #2: Range and s.d. for Spectral Energy

Energy in entire band



RMS Energy in 20 ms windows; range and standard deviation of these values

LTAS Measures #3: Pairwise Change



Results



to track how
every actually
fluctuates across
the data.

Analysis	Classification Accuracy	Most Predictive Variable(s)
Linguistic Measures	76.9% (with 6 variables)	percV
Linguistic Measures (top 3 variables only)	70.3%	percV, Duration, varcoV

← traditional/
rhythmic
methods.

Analysis	Classification Accuracy	Most Predictive Variable(s)
LTAS Measures	100% (with 24 variables)	Rng1000
LTAS Measures (top 3 variables only)	83.5%	Rng1000, PV8000, N2RMS500

they
are much
more
powerful.

Chance = 33.3%

EMS & LTAS

- There are many aspects that can be changed and many of the original choices were arbitrary:

- Length of input
- Frequency bands
- Dependent Variables
- Analysis Windows
- Restricting EMS to under 10 Hz

— you can design how you want

Loop to the researchers.